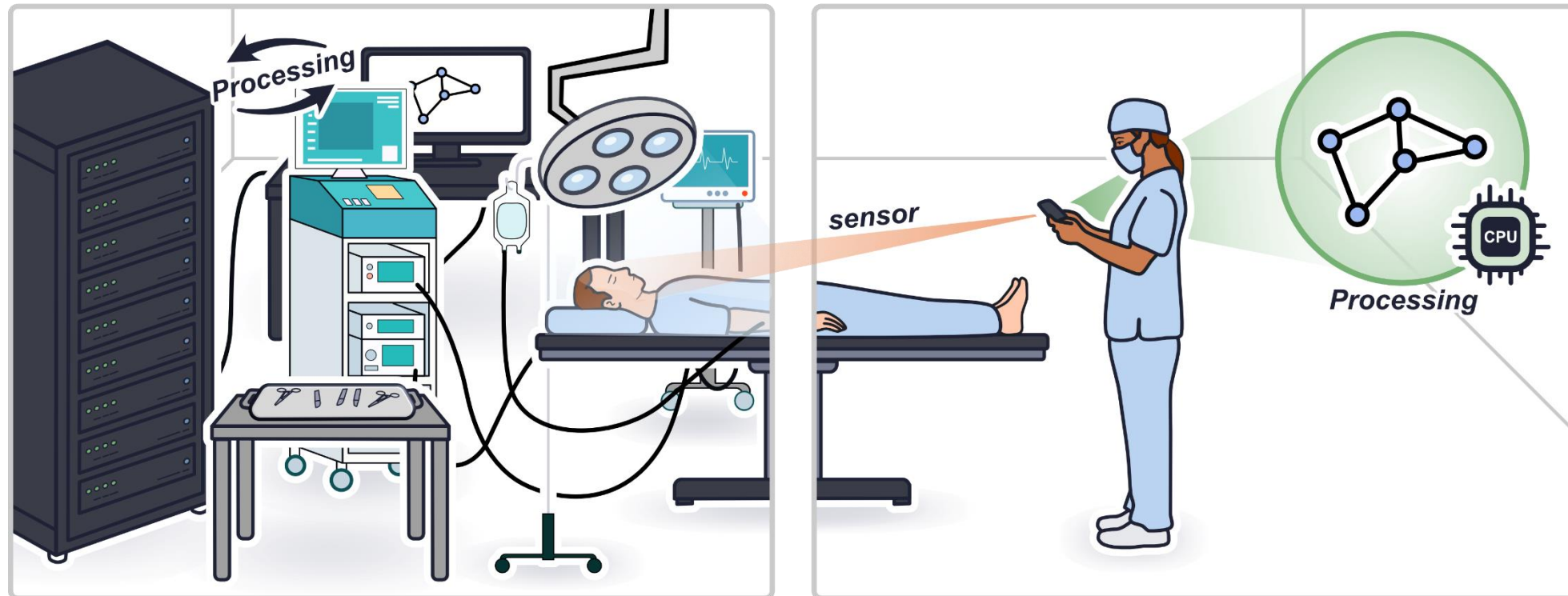
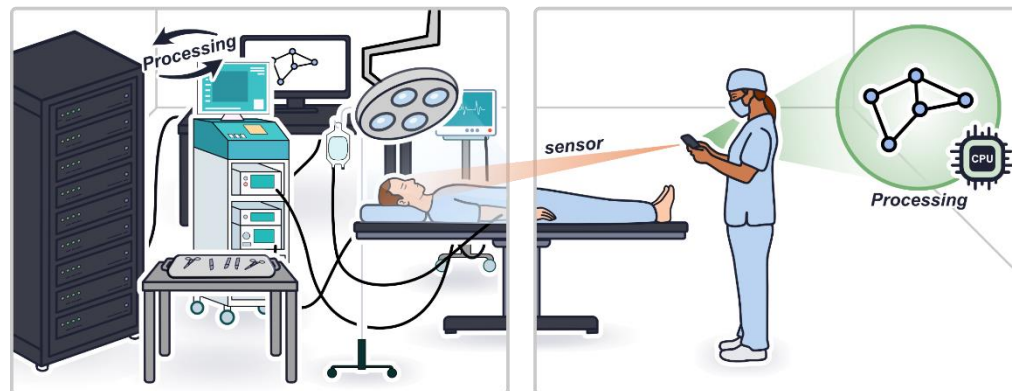


Sterilizable Scene Graph Generation for Operating Rooms

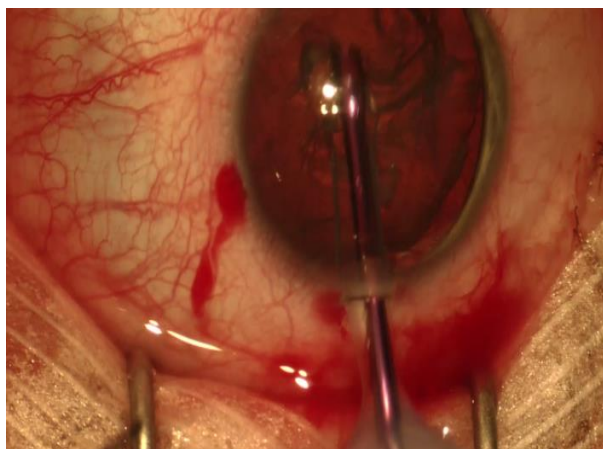


Nick Lemke, Ssharvien Kumar Sivakumar, Antoine P. Sanner, John Kalkhof,
Henry John Krumb, Ghazal Ghazaei, and Anirban Mukhopadhyay

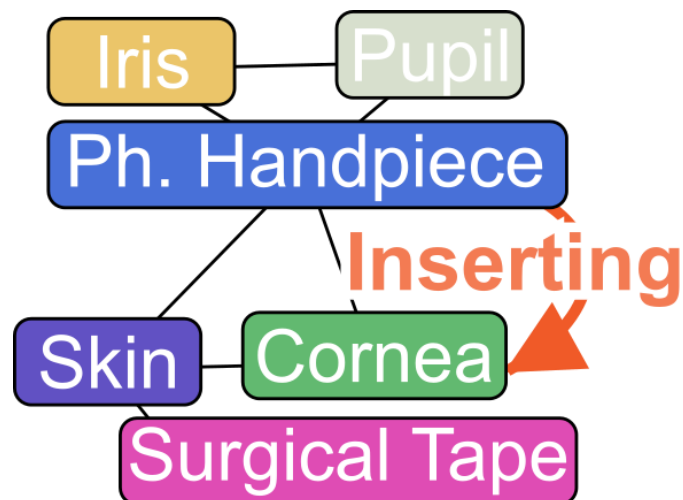
Introduction



TECHNISCHE
UNIVERSITÄT
DARMSTADT



Input Sequence

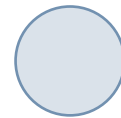


Scene Graph

The Phacoemulsification Handpiece **inserts** into the cornea.

Caption

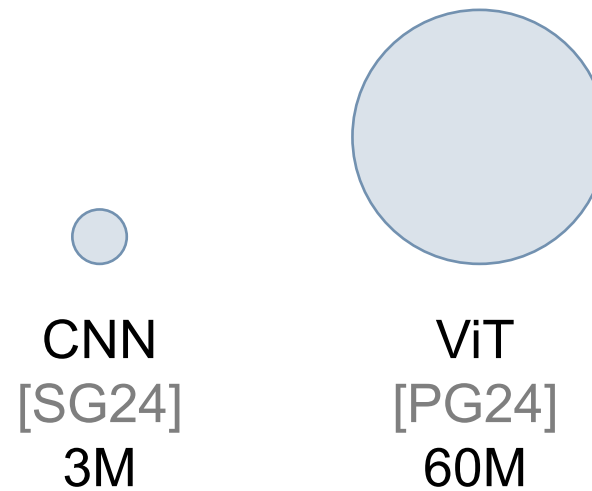
Introduction



CNN
[SG24]
3M

[SG24] Sanner, A. P., Grauhan, N. F., Brockmann, M. A., Othman, A. E., & Mukhopadhyay, A. Voxel scene graph for intracranial hemorrhage. MICCAI 2024

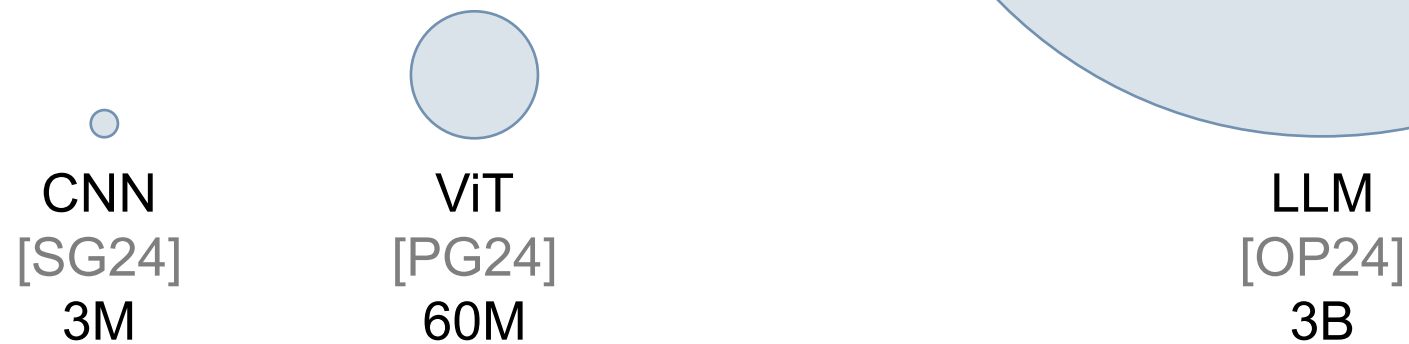
Introduction



[SG24] Sanner, A. P., Grauhan, N. F., Brockmann, M. A., Othman, A. E., & Mukhopadhyay, A. Voxel scene graph for intracranial hemorrhage. MICCAI 2024

[PG24] Pei, J., Guo, D., Zhang, J., Lin, M., Jin, Y., & Heng, P. A. S²Former-OR: Single-Stage Bi-Modal Transformer for Scene Graph Generation in OR. TMI 2024

Introduction

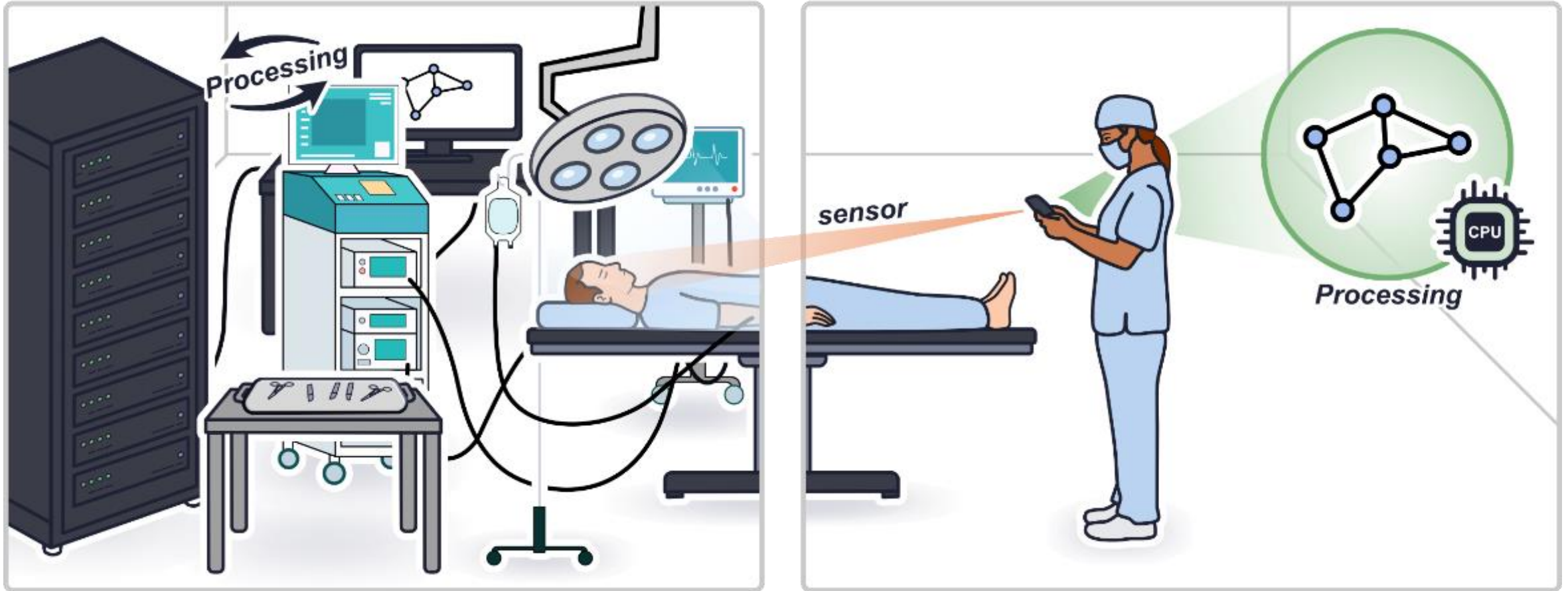


[SG24] Sanner, A. P., Grauhan, N. F., Brockmann, M. A., Othman, A. E., & Mukhopadhyay, A. Voxel scene graph for intracranial hemorrhage. MICCAI 2024

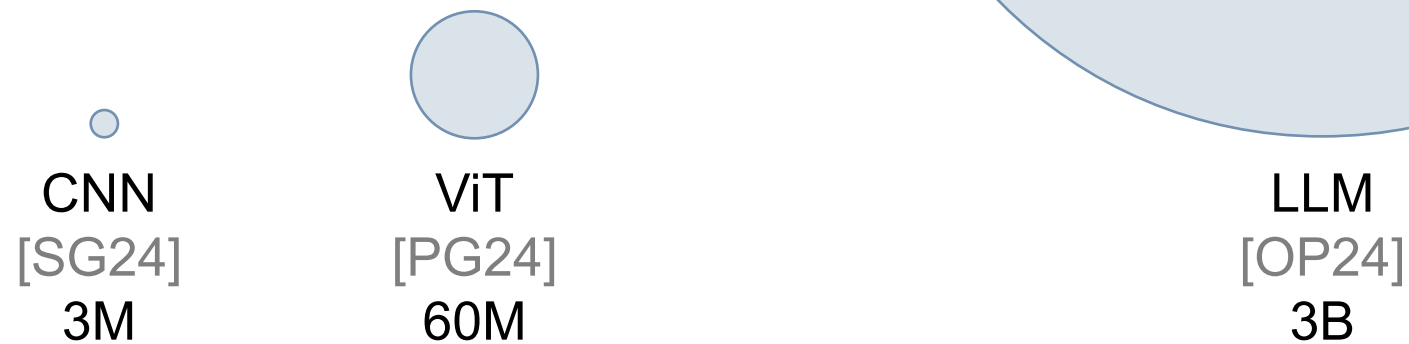
[PG24] Pei, J., Guo, D., Zhang, J., Lin, M., Jin, Y., & Heng, P. A. S²Former-OR: Single-Stage Bi-Modal Transformer for Scene Graph Generation in OR. TMI 2024

[OP24] Özsoy, E., Pellegrini, C., Keicher, M., & Navab, N. Oracle: Large vision-language models for knowledge-guided holistic or domain modeling. MICCAI 2024

Introduction



Introduction

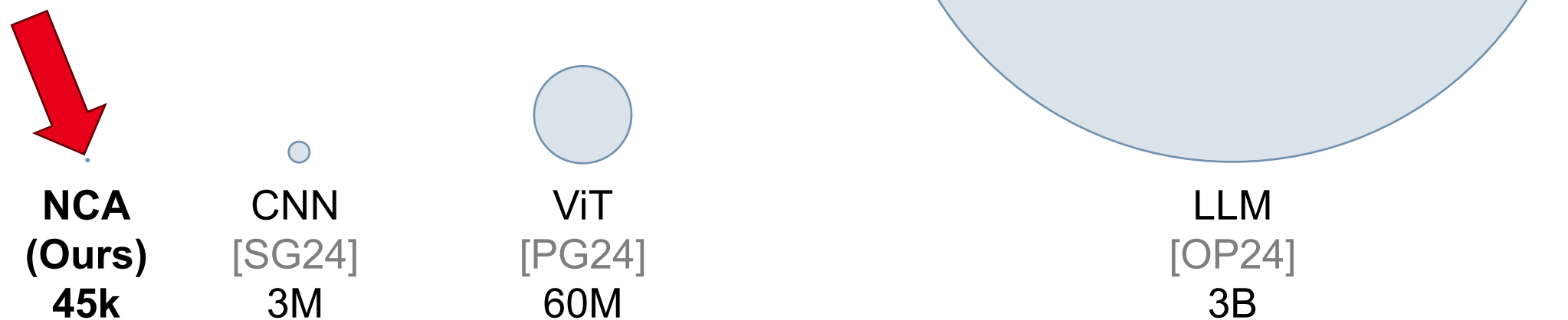


[SG24] Sanner, A. P., Grauhan, N. F., Brockmann, M. A., Othman, A. E., & Mukhopadhyay, A. Voxel scene graph for intracranial hemorrhage. MICCAI 2024

[PG24] Pei, J., Guo, D., Zhang, J., Lin, M., Jin, Y., & Heng, P. A. S²Former-OR: Single-Stage Bi-Modal Transformer for Scene Graph Generation in OR. TMI 2024

[OP24] Özsoy, E., Pellegrini, C., Keicher, M., & Navab, N. Oracle: Large vision-language models for knowledge-guided holistic or domain modeling. MICCAI 2024

Introduction

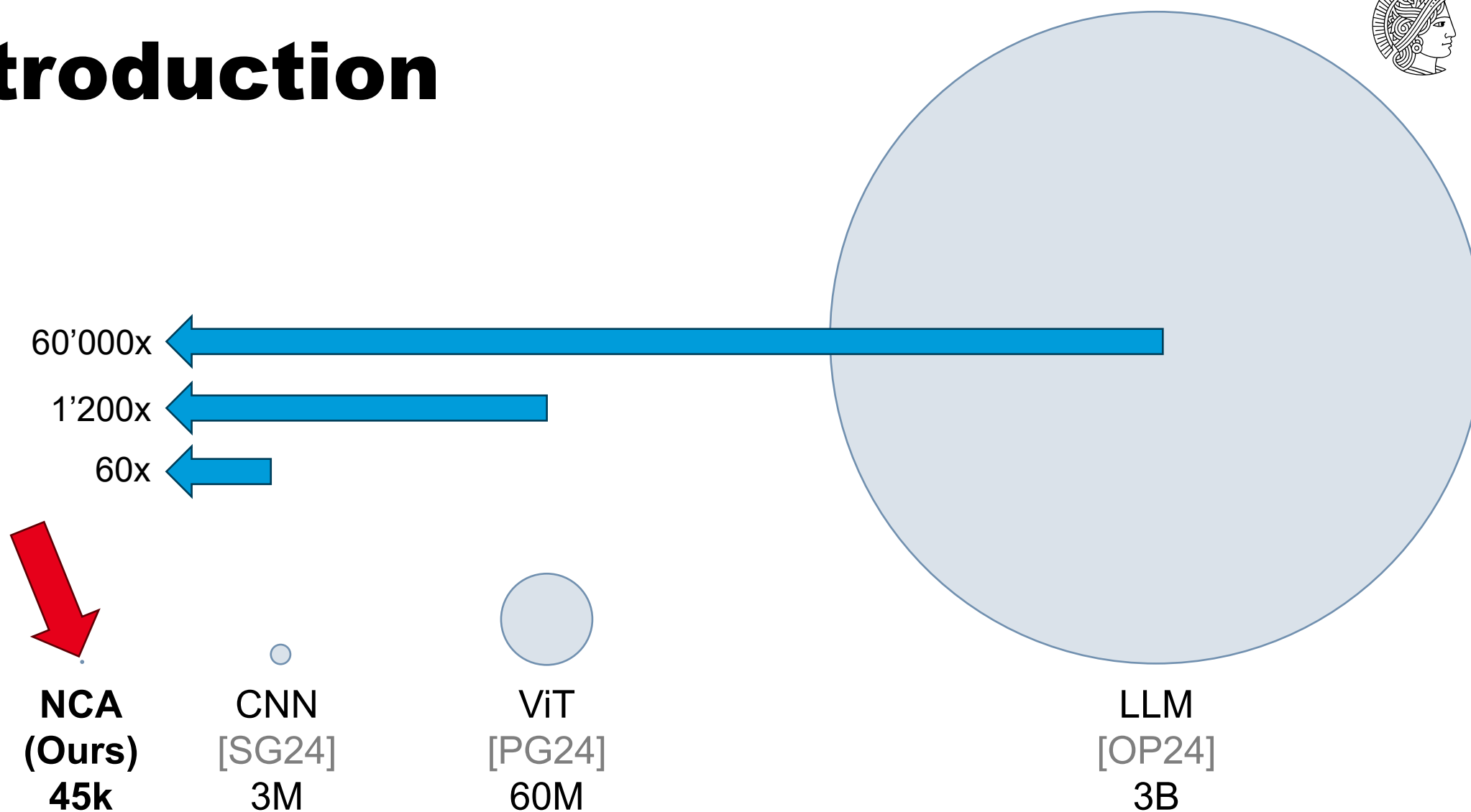


[SG24] Sanner, A. P., Grauhan, N. F., Brockmann, M. A., Othman, A. E., & Mukhopadhyay, A. Voxel scene graph for intracranial hemorrhage. MICCAI 2024

[PG24] Pei, J., Guo, D., Zhang, J., Lin, M., Jin, Y., & Heng, P. A. S²Former-OR: Single-Stage Bi-Modal Transformer for Scene Graph Generation in OR. TMI 2024

[OP24] Özsoy, E., Pellegrini, C., Keicher, M., & Navab, N. Oracle: Large vision-language models for knowledge-guided holistic or domain modeling. MICCAI 2024

Introduction

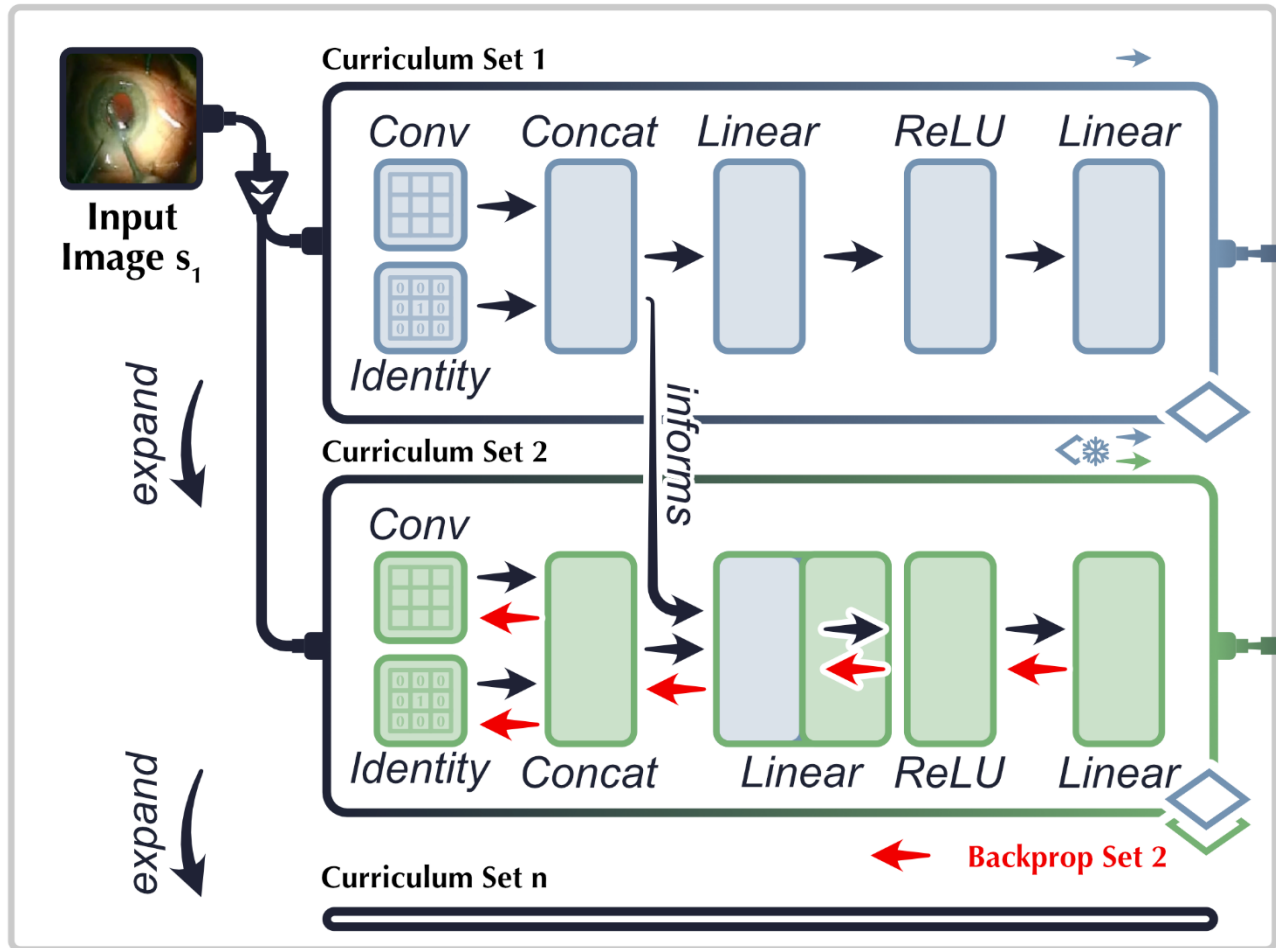


[SG24] Sanner, A. P., Grauhan, N. F., Brockmann, M. A., Othman, A. E., & Mukhopadhyay, A. Voxel scene graph for intracranial hemorrhage. MICCAI 2024

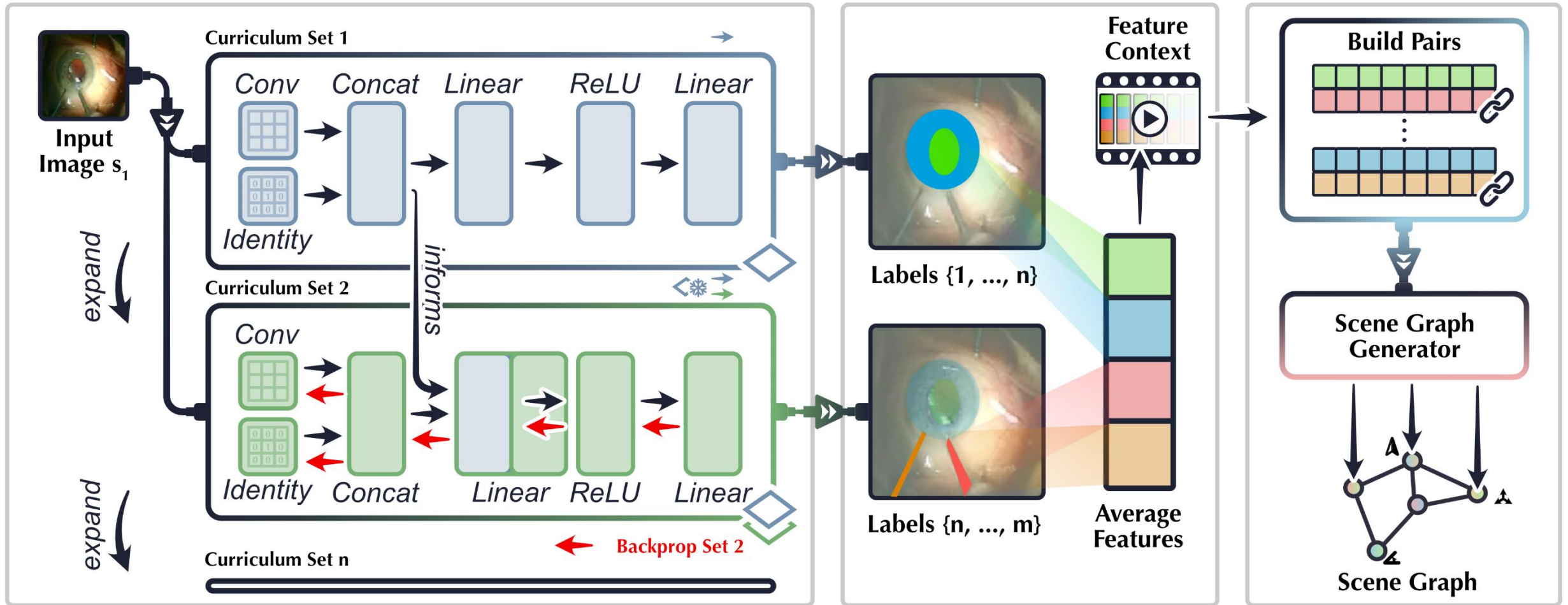
[PG24] Pei, J., Guo, D., Zhang, J., Lin, M., Jin, Y., & Heng, P. A. S²Former-OR: Single-Stage Bi-Modal Transformer for Scene Graph Generation in OR. TMI 2024

[OP24] Özsoy, E., Pellegrini, C., Keicher, M., & Navab, N. Oracle: Large vision-language models for knowledge-guided holistic or domain modeling. MICCAI 2024

SG-NCA

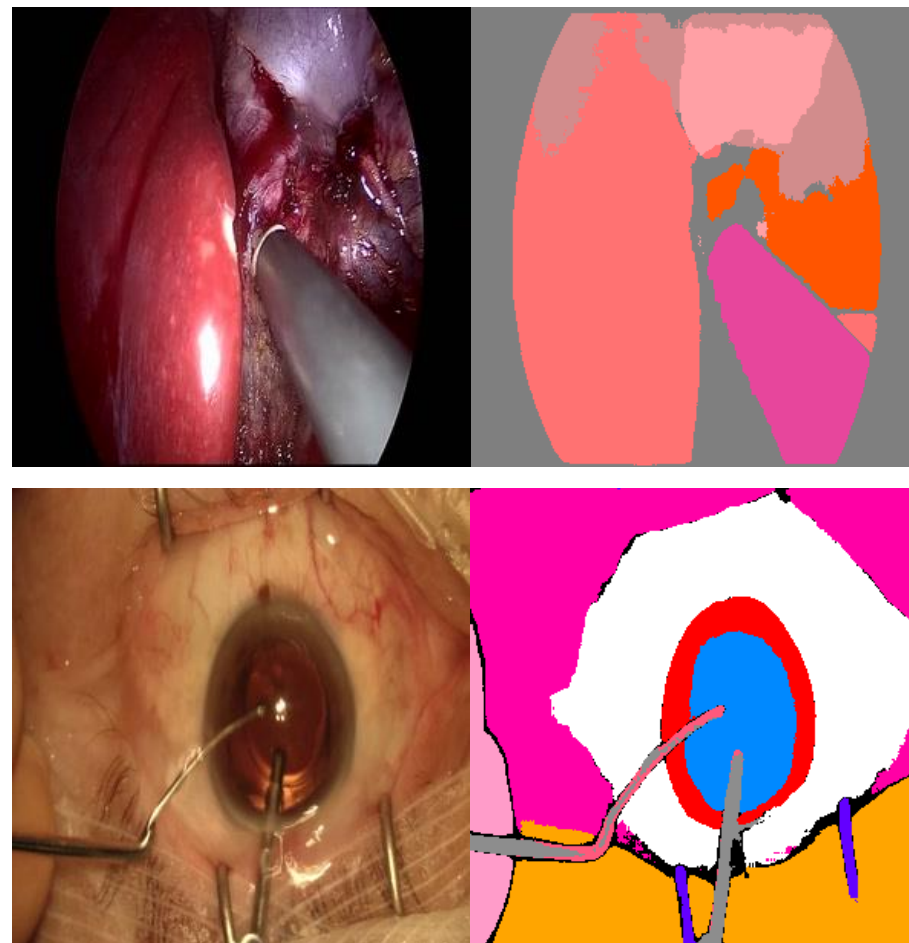


SG-NCA



Evaluation: Data

- Cholec80 [TS16], CATARACTS [AL19]
- CholecT50 [NG20], CAT-SG [HG25]
- SASVi Pseudo-Seg [SF25]
- CholecSeg8k [HK20], CaDIS [GF21]



[AL19] Al Hajj, H., Lamard, M., Conze, P. H., Roychowdhury, S., Hu, X., Maršalkaitė, G., ... & Quellec, G. CATARACTS: Challenge on automatic tool annotation for cataRACT surgery. MedIA 2019

[TS16] Twinanda, A. P., Shehata, S., Mutter, D., Marescaux, J., De Mathelin, M., & Padoy, N. Endonet: a deep architecture for recognition tasks on laparoscopic videos. TMI 2016

[NG20] Nwoye, C. I., Gonzalez, C., Yu, T., Mascagni, P., Mutter, D., Marescaux, J., & Padoy, N. Recognition of instrument-tissue interactions in endoscopic videos via action triplets. MICCAI 2020

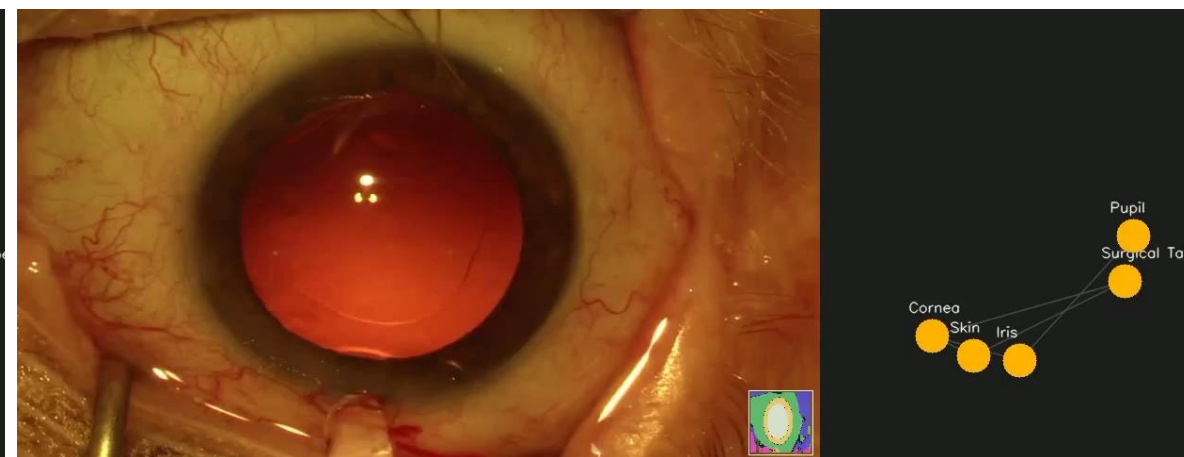
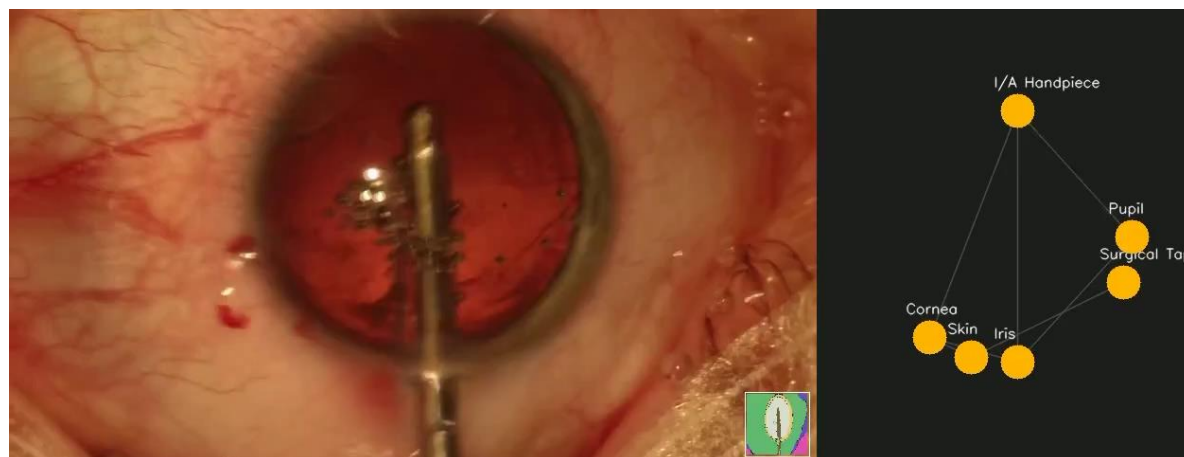
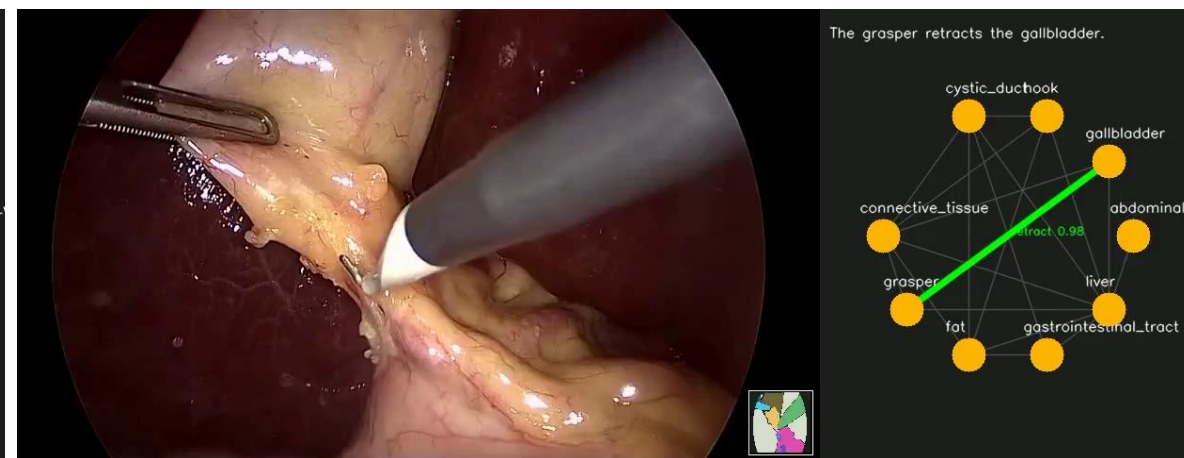
[HG25] Holm, F., Ünver, G., Ghazaei, G., & Navab, N. Cat-sg: A large dynamic scene graph dataset for fine-grained understanding of cataract surgery. MICCAI 2025

[SF25] Sivakumar, S. K., Frisch, Y., Ranem, A., & Mukhopadhyay, A. SASVi: segment any surgical video. IPCAI 2025

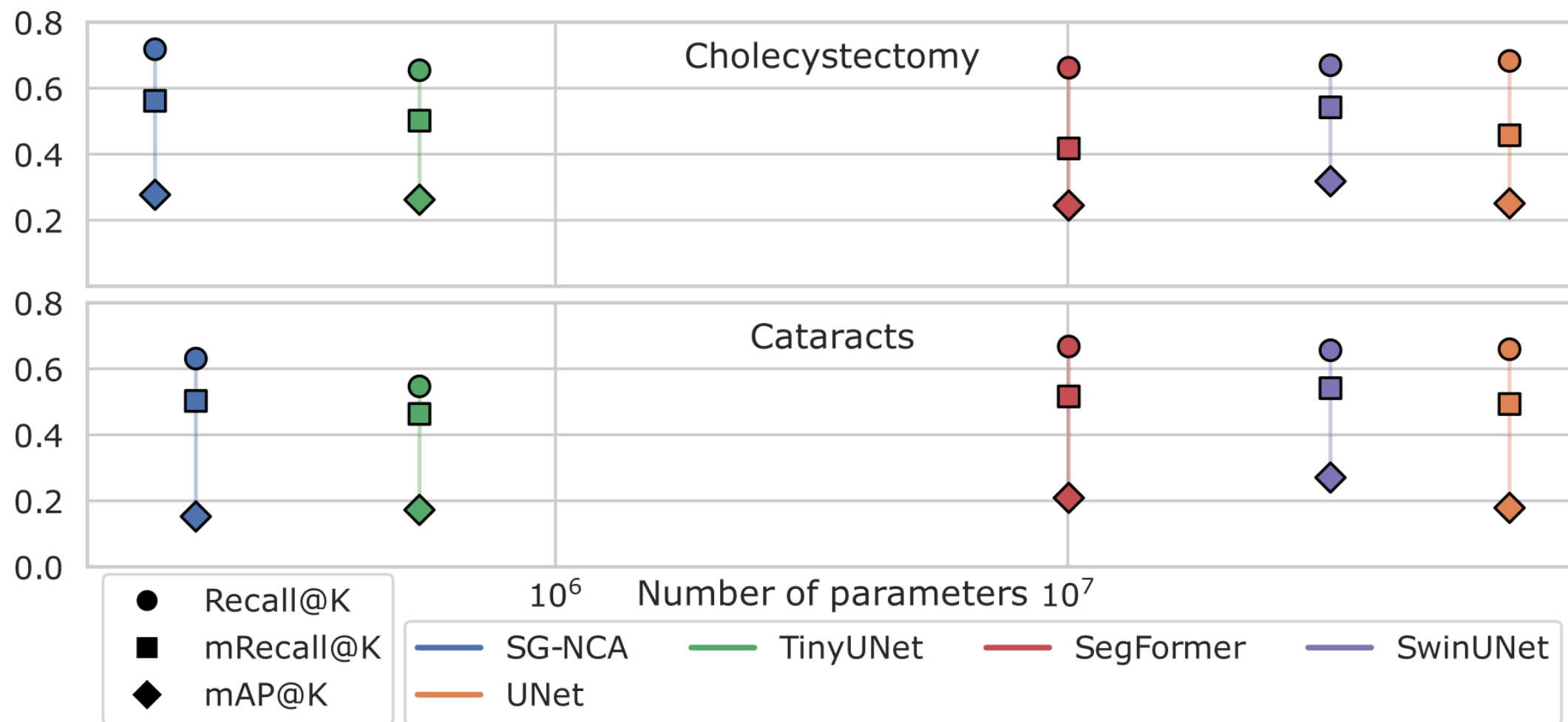
[HK20] Hong, W. Y., Kao, C. L., Kuo, Y. H., Wang, J. R., Chang, W. L., & Shih, C. S. Cholecseg8k: a semantic segmentation dataset for laparoscopic cholecystectomy based on cholec80. arXiv 2020

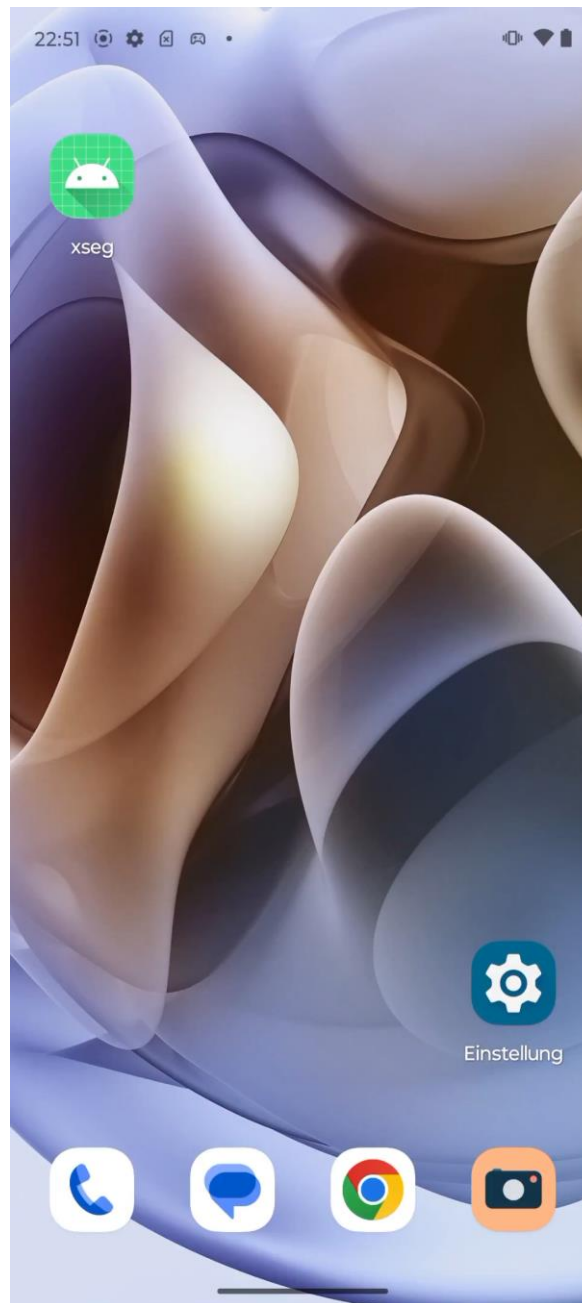
[GF21] Grammatikopoulou, M., Flouty, E., Kadkhodamohammadi, A., Quellec, G., Chow, A., Nehme, J., ... & Stoyanov, D. CaDIS: Cataract dataset for surgical RGB-image segmentation. MedIA 2021

Evaluation: Scene Graphs



Evaluation: Comparison





Conclusion

- Curriculum training scales SG-NCA for multiple classes
- SG-NCA performs on-par with much larger baselines
- Deployment on edge devices is feasible



Code



Preprint



Nick Lemke